

Informatique Embarquée

Colle TP

1. Ecrire les commandes Linux nécessaires pour accéder au dossier contenant les GPIO.
2. Ecrire les commandes Linux nécessaires pour accéder au PIOB16.
3. Sur quelles broches des deux connecteurs J6 ou J7 se trouvent les PIOB16, PIOB17, PIOB18 et PIOB19 ?
4. Utiliser les quatre broches pour réaliser un compteur impair contrôlé par un switch, dont le résultat sera affiché sur l'afficher 7 segments de la maquette.

Pin #	ID	I/O line	Alt	Description	Pin #	ID	Line	Alt	Description
J7.1		GND		Signal ground	J6.1		3.3V		3.3 volt DC power line
J7.2		GND		Signal ground	J6.2		3.3V		3.3 volt DC power line
J7.3	50	PB18		General purpose I/O	J6.3	60	PB28	RTS1	Request to send ttyS2
J7.4	51	PB19		General purpose I/O	J6.4	39	PB7	RXD1	Receive data ttyS2
J7.5	48	PB16		General purpose I/O	J6.5	38	PB6	TXD1	Transmit line ttyS2
J7.6	49	PB17		General purpose I/O	J6.6	61	PB29	CTS1	Clear to send ttyS2
J7.7	34	PB2		General purpose I/O	J6.7	58	PB26	RTS0	Request to send ttyS1
J7.8	35	PB3		General purpose I/O	J6.8	37	PB5	RXD0	Receive data ttyS1
J7.9	32	PB0		General purpose I/O	J6.9	36	PB4	TXD0	Transmit data ttyS1
J7.10	33	PB1		General purpose I/O	J6.10	52	PB27	CTS0	Clear to send ttyS1
J7.11	78	PC14		General purpose I/O 1.8V	J6.11				N.C. (See the schematic)
J7.12	79	PC15		General purpose I/O 1.8V	J6.12		5V		5 volt DC power line
J7.13	76	PC12		General purpose I/O 1.8V	J6.13	43	PB11	RXD3	Receive data ttyS4
J7.14	77	PC13		General purpose I/O 1.8V	J6.14	42	PB10	TXD3	Transmit data ttyS4
J7.15	73	PC9		General purpose I/O 1.8V	J6.15	45	PB13	RXD5	Receive data ttyS6
J7.16	74	PC10	CTS3	Clear to send ttyS4 1.8V	J6.16	44	PB12	TXD5	Transmit data ttyS6
J7.17	71	PC7		Red led line 1.8V	J6.17	53	PB21		General purpose I/O
J7.18	72	PC8	RTS3	Request to send ttyS4 1.8V	J6.18	52	PB20		General purpose I/O
J7.19	69	PC5		General purpose I/O 1.8V	J6.19	63	PB31		General purpose I/O
J7.20	70	PC6		Reserved line. Used to read if the client USB port is wired to a PC	J6.20	62	PB30		General purpose I/O
J7.21	41	PB9	RXD2	Receive data ttyS3	J6.21	31	PA31	TXD4	Transmit data ttyS5
J7.22	40	PB8	TXD2	Transmit data ttyS3	J6.22	30	PA30	RXD4	Receive data ttyS5
J7.23		BATT		RTC Battery input	J6.23				N.C. (See the schematic)
J7.24		PGD		Power good NetusPS1	J6.24	6	PA6		General purpose I/O
J7.25		POK		3.3V NetusPS1 aux output stable	J6.25	7	PA7		General purpose I/O
J7.26		SHDNPS#		PS1 shutdown. Active low	J6.26	9	PA9		General purpose I/O
J7.27		NRST		Reset output	J6.27	67	PC3	AD3	Analog input 3
J7.28		SHDN#		Turn off the CPU when low	J6.28	66	PC2	AD2	Analog input 2
J7.29		5V		5 volt power line	J6.29	65	PC1	AD1	Analog input 1
J7.30		WAKEUP		Wake up input	J6.30	64	PC0	AD0	Analog input 0
J7.31	55	PB23	DCD0	Data carrier detect ttyS1	J6.31	24	PA24	SCL	I2C Clock
J7.32	54	PB22	DSR0	Data set ready ttyS1	J6.32	23	PA23	SDA	I2C Data
J7.33	57	PB25	RIO	Ring indicator ttyS1	J6.33		AVDD		Clean 3.3V out for A/D circuit
J7.34	56	PB24	DTR0	Data terminal ready ttyS1	J6.34		VREF		A/D voltage reference input
J7.35	28	PA28		General purpose I/O	J6.35		AGND		Analog ground
J7.36	27	PA27		General purpose I/O	J6.36	10	PA10		General purpose I/O
J7.37	26	PA26		General purpose I/O	J6.37	22	PA22		General purpose I/O
J7.38	25	PA25		General purpose I/O	J6.38	11	PA11		General purpose I/O
J7.39		3.3V		3.3 volt power line	J6.39		GND		Signal ground
J7.40		3.3V		3.3 volt power line	J6.40		GND		Signal ground